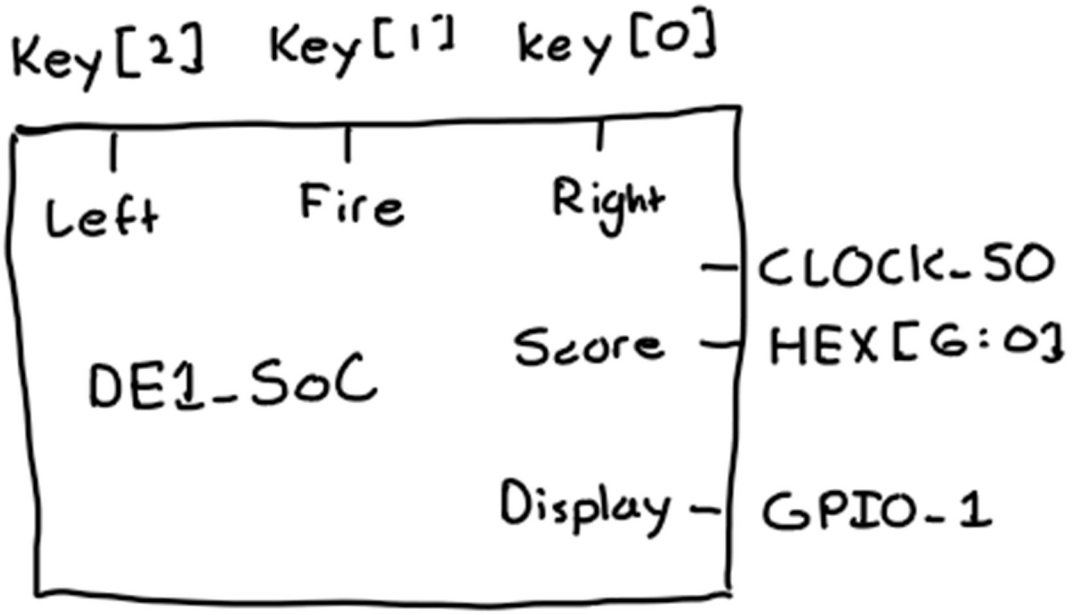
Aaryan Pawar

Final Project Proposal

I developed a simplified version of the classic arcade game *Space Invaders*, featuring interactive gameplay and basic game logic. In this game, the player controls a green-colored character located at the bottom of the screen. The character can move horizontally (left and right) and shoot bullets vertically upwards to eliminate incoming enemies (aliens).

The alien enemies are displayed in red and are programmed to move horizontally across the screen in a pseudo-random pattern. This randomness is generated using a Linear Feedback Shift Register (LFSR), which ensures that both the movement and firing patterns of the aliens are unpredictable, enhancing replayability and challenge. In addition to moving, the aliens can shoot orange-colored bullets downward at the player, also at pseudo-random intervals determined by the LFSR.

When a player's green bullet hits a red alien, the alien is removed from the screen, and the player earns one point. The point total is tracked using an unsigned integer counter displayed on the screen. Alien bullets can potentially hit the player, introducing a risk-reward element and encouraging strategic movement and shooting.

This implementation features basic enemy AI driven by an LFSR for randomized behavior, a simple collision detection system between player bullets and aliens, and between alien bullets and the player. It also includes real-time controls for horizontal movement and firing, visual color-coded distinctions (player in green, aliens in red, alien bullets in orange), and a scoring system based on successful hits.

Block Diagram